MILITARY SPECIFICATION
CARTRIDGE, 7.62X39MM, BALL

1. SCOPE

1.1 This specification covers Cartridge, 7.62X39MM, Ball intended for use in automatic rifles chambered for that cartridge.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Federal
Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

SPECIFICATIONS
Military

MIL-I-45607	- Inspection Equipment, Acquisition Maintenance and Disposition of

DRAWINGS
U.S. Army Munitions Command

C11731648	- Cartridge, 7.62X39MM, Ball
F11738715	- Packing and Marking 7.62X39MM, Box, Shipping
C7643674	- Classification of Cartridge Case Defects

PUBLICATIONS

AMCR 715-505 Volume 3	- Ammunition Ballistic Acceptance Test Methods, Volume 3, Test Procedures for 7.62MM Cartridge

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2.1 (Cont'd)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General - The cartridge shall comply with Drawing (Dwg) C11731648, referenced specifications and the following:

3.2 Bullet extraction - The force required to extract the bullet from the cartridge case shall be not less than 35 pounds.

3.3 Primer sensitivity - The energy imparted by a 4-ounce steel ball falling 16 inches shall cause initiation of the primer. The energy imparted by a 4-ounce steel ball falling 3 inches shall not cause initiation of the primer.

3.4 Residual stress - The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.5 Waterproof - The cartridge shall not release more than one bubble of air when subjected to a pressure differential of 5 pounds per square inch (psi) for 15 seconds.

3.6 Accuracy - The average of the mean radii of all targets of the sample cartridges, fired in standard accuracy test weapons over a range of 200 yards, shall not exceed 5.0 inches.

3.7 Action time - The action time (overall primer ignition, propellant burning and bullet-barrel time) of the cartridge shall not exceed 4 milliseconds.

3.8 Velocity - The average velocity of the sample cartridges, conditioned at 68° to 72° Fahrenheit (F), shall be 2275 feet per second (ft/sec) plus or minus 30 ft/sec at 78 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 32 ft/sec.

3.9 Chamber pressure - The average chamber pressure of the sample cartridges conditioned and fired at 68° to 72° F shall not exceed 45,000 psi. The average chamber pressure of the sample cartridges conditioned and fired at -65° F and at 125° F shall not exceed 48,000 psi.

3.10 Function and casualty - The cartridge shall function without casualty at ambient temperature and at -65° and 125°F.

3.11 Stripping - The jacket of the bullet, or any part thereof, shall not strip from the slug when the cartridge is fired.

3.12 Workmanship - The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

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3.12.1 Metal defects - The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.12.2 Foreign matter - The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions - Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample

4.2.1 Initial production sample - At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 3500 cartridges and 500 primed cases. In addition, 100 bullets shall be submitted for visual and dimensional examinations. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test - After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure - Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval. Determination as to acceptability of any initial production sample shall be based upon results of initial tests only, and no second tests shall be permitted on that initial production sample.

4.3 Inspection provisions

4.3.1 Lot

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4.3.1.1 Submission of product - The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification - Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination - One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The Acceptable Quality Level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent. All non-conforming cartridges shall be rejected.

4.3.2.1 Classification of defects - The classification of defects shall be as specified in Table I.

Table I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Cartridge				
1	Discolored, dirty, oily, smeared			X	
2	Corroded or stained, if etched		X		
	Case				
4	Round Head		X		
5	Dent				X
6	Split Case				
	in K, L, or M location	X			
	in I, S or J location		X		
7	Perforated case	X			
8	Draw Scratch				X
9	Scratch			X	
10	Beveled underside of head		X		
11	Case mouth not crimped in cannelure		X		
12	Scaly metal				X
13	No chamfer on head (rim)		X		
14	Fold			X	
15	Wrinkle			X	
16	Buckle			X	
17	Bulge			X	
18	Presence of head stamp		X		

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Table I (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
19	Defective head			X	
20	Defective mouth			X	
21	No visible evidence of mouth anneal		X		
	Bullet				
22	Dent			X	
23	Scratch			X	
24	Split bullet jacket		X		
25	Loose bullet		X		
26	Missing cannelure		X		
27	Scaly metal (bullet)				X
28	Upset (crooked) point			X	
30	Blunt point			X	
31	Defective cannelure			X	
	Primer				
32	No primer	X			
33	Cocked primer	X			
34	Inverted primer	X			
35	Loose primer		X		
36	Nicked or dented primer			X	
37	No waterproofing material (primer pocket joint)			X	
38	Defective crimp			X	
	Gaging				
39	Total length		X		
40	Cartridge profile failure (requiring more than 20 lbs dead weight to insert in profile and alignment gage)		X		
41	Diameter of extractor groove, max.		X		
42	Diameter of extractor groove, min.			X	
43	Diameter of head		X		
44	Thickness of head		X		
45	Length to shoulder datum		X		
46	Depth of primer		X		
	Weighing				
47	Weight, min. 2/	X			

1/ Refer to MIL-STD-636 (NATO caliber 7.62mm section) for visual defect standard for defects 1 through 38, with the exception of defect 18.

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2/ Each lightweight cartridge shall be disassembled and the propellant weighed. Each such cartridge found to contain 10 grains or more of propellant shall be classed as a major defect. Any cartridge containing less than 10 grains of propellant shall be classed as a critical defect.

4.3.3 Tests - The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples - The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

Table II

Test	Number of Cartridges				Requirement Para.
	Ambient Temp.	70° ± 2°F	Low Temp (See 4.4.9.1)	High Temp (See 4.4.9.2)	
Bullet extraction 1/	25				3.2
Primer sensitivity 2/					3.3
Residual Stress (Mercurous nitrate)1/	50				3.4
Waterproof 3/	50				3.5
Accuracy 4/	90				3.6
Action time 1/		50			3.7
Velocity 4/		20			3.8
Chamber pressure 5/		20	20	20	3.9
Function & Casualty 6/	480		480	480	3.10
Stripping 7/					3.11
Hardness Head 1/	10				Drawing

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ Prior to assembly into cartridges, a sample of primers from each primer lot shall be inserted into cartridge cases and tested for sensitivity. The cartridge cases for this test shall be taken from regular production. The test sample shall consist of the number of primed cartridge cases required

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2/ (Cont'd)

for performance of a complete run-down test wherein fifty primers are tested at each height. If the average critical height ($\bar{H}$) plus five standard deviations (5 sigma) exceeds 16 inches, or if the average critical height ($\bar{H}$) minus two standard deviations (2 sigma) is less than 3 inches, the lot of primers shall be rejected subject to performance of a second complete run-down test wherein one-hundred primers are tested at each height. Failure of the second sample to comply with the above criteria shall be cause for rejection of the lot of primers.

3/ Failure of ten or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than three but less than ten cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if in the combined first and second sample, ten or more cartridges fail to comply with the applicable requirement.

4/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

5/ Failure of the cartridges in any sample to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test for the temperature or temperatures at which the failure occurred. Failure of the cartridges of the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

6/ The lot shall be rejected when function and casualty defects plus firing defects observed in all other firing tests exceed the acceptance number for the cumulative sample of Table III. If the number of defects found in the first tests exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantities specified under function and casualty test, shall be fired in the service weapon specified therefor. This procedure shall apply regardless of the weapon or weapons in which the firing defects occurred in the first test. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

7/ This test shall be conducted on the initial production sample only. Determination of compliance with the bullet stripping requirement shall be made during the function and casualty test.

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4.3.3.2 Firing defects - Firing defects and acceptance numbers shall be as specified in Table III.

Table III

Defects	Acceptance First Sample	Acceptance Cumulative (1st & 2nd Sample)
1. Misfire 6/		
a. No vent hole	0	1
b. Other	1	2
2. Bullet remaining in bore 1/	0	-
3. Primer leak:		
a. Perforation in firing pin indent in primer cup 2/	0	-
b. Escape of gas through primer cup other than a. above	1	2
c. Escape of gas around primer cup 5/	44	105
d. Blown primer or primer falls out of pocket on retraction of bolt	0	1
e. Primer remains in pocket but is physically loose	1	2
4. Case casualties		
a. Longitudinal split 3/		
(1) Neck & shoulder (I or S)	28	66
(2) Body (J)	2	4
(3) Body (K)	0	1
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 3/		
(1) Partial, shoulder or body (J & S)	1	2
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1
5. Failure to extract	0	1
6. Weapons stoppage 4/	0	1

1/ No second sample permitted. Lot shall be rejected.

2/ If one or more defects are found in the first sample, a second sample consisting of double the quantity of cartridges specified under Function and Casualty of Table II shall be fired. Prior to the testing of the second sample, the firing pin of the specific rifle(s) in which the defect originally occurred shall be replaced by a new firing pin. If an additional primer perforation is found in the second sample, the lot shall be rejected.

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4.3.3.2 (Cont'd)

3/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

4/ All stoppages attributable to the ammunition, with the exception of misfire, complete rupture or failure to extract, observed in all tests shall be included.

5/ Of the cartridges in which this defect occurs, not more than 14 in the first sample not more than 30 in the cumulative sample shall exhibit escape of gas around 50% or more than 50% or the periphery.

6/ Each cartridge that misfires shall be disassembled and examined for determination of cause for misfire.

4.3.4 Packaging, packing and marking inspection - During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a blank cartridge shall be classed as a critical defect. Occurrence of any other type shall be classed as a major defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

4.3.5 Inspection equipment - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.4 Test methods and procedures

4.4.1 Bullet extraction - The test shall be conducted in accordance with AMCR 715-505, Volume 3. The rate of travel of the test head shall be not less than three nor more than six inches per minute.

4.4.2 Primer sensitivity - The primers shall be inserted into cartridge cases to the depth specified on Drawing C11731648. The test shall be conducted in accordance with AMCR 715-505, Volume 3, except that the headspace shall be 1.247 + .003 inch.

4.4.3 Residual stress - (mercurous nitrate) - The test shall be conducted in accordance with AMCR 715-505, Volume 3.

4.4.4 Waterproof - The test shall be conducted in accordance with AMCR 715-505, Vol. 3.

4.4.5 Accuracy - The test shall be conducted in accordance with AMCR 715-505, Volume 3, except that the headspace shall be 1.247 + .003 inch.

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4.4.6 Action time - The test shall be conducted in accordance with AMCR 715-505, Volume 3.

4.4.7 Velocity - The test shall be conducted in accordance with AMCR 715-505, Volume 3.

4.4.8 Chamber pressure - The tests shall be conducted in accordance with AMCR 715-505, Volume 3, and 4.4.9.

4.4.9 Temperature - The tests shall be conducted in accordance with AMCR 715-505, Volume 3. The weapon in which these tests are fired shall be at room temperature.

4.4.9.1 Low temperature test - The test sample shall be stored at minus 65 degrees F, plus or minus 5 degrees, for not less than 6 hours and shall be fired at that temperature.

4.4.9.2 High temperature test - The test sample shall be stored at 125 degrees F, plus or minus 2 degrees, for not less than 4 hours and shall be fired at that temperature.

4.4.10 Function and casualty - The tests shall be conducted in accordance with AMCR 715-505, Volume 3 and 4.4.8. Two rifles shall be used and the following number of cartridges be fired in each;

	First Sample	Second Sample (if required)
Ambient temperature	240	480
Low temperature	240	480
High temperature	240	480

4.4.11 Stripping - The test shall be conducted in accordance with AMCR 715-505, Volume 3,

4.4.12 Hardness testing - The bullets shall be extracted, the propellant removed and the primers extracted. Each cartridge case of the sample shall be prepared and placed on the appropriate test fixture for testing in accordance with Federal Test Method Standard No. 151, Method 244.1.

4.4.12.1 Case head - The individual hardness value for each prescribed point on the head section of each sample case shall be recorded. Any value failing to meet the drawing requirement at a prescribed point(s) shall be cause for remeasurement of hardness at the corresponding point(s) on the opposite side of the primer pocket of the same head section from which the initial value was obtained. The higher of the two measurements shall be recorded as the value of record for determination of conformance to drawing requirements.

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4.4.13 Defect penalty - In any ballistic test, except function and casulty, in which the occurrence of a firing defect prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the acceptance or initial production sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level A (World-Wide Shipment)- The cartridges shall be packed in accordance with Drawing F11738715.

5.2 Marking and labeling - Packing boxes shall be marked and labeled in accordance with the applicable drawing cited in 5.1 or as directed by the contracting officer.

6. NOTES

6.1 Ordering data - Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.5 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirement.

6.2 Hazard notice - The cartridge described herein and certain of its components are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

Custodian:

Army - MU

Preparing activity:

Army - MU (FA)

Project No. 1305 - A683

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